

BATT : 7.2~24V

VBUS ← PWR_FLAG

+5V max current : 5A

+3.3V max current : 1A

VBUS

+5V

+3.3V

GND

Legend:

- 0 0 0 10BT Half-duplex, Auto-negotiation disabled
- 0 0 1 10BT Full-duplex, Auto-negotiation disabled
- 0 1 0 100BT Half-duplex, Auto-negotiation disabled
- 0 1 1 100BT Full-duplex, Auto-negotiation disabled
- 1 0 0 100BT Half-duplex, Auto-negotiation enabled
- 1 0 1 Not used
- 1 1 0 Not used
- 1 1 1 All capable, Auto-negotiation enabled

The diagram illustrates the electrical connection between a USB-C receptacle (J10) and an FT232RL module (U11).

USB-C Connector (J10) Details:

- VBUS:** Connected to pin A4. A 0.1uF capacitor (C27) is connected between VBUS and GND, and a 4.7uF capacitor (C41) is connected between VBUS and GND.
- CC1, CC2:** Connected to pins A5 and B5 respectively.
- D-:** Connected to pin A7. A 4.7k resistor (R27) is connected between D- and GND.
- D+:** Connected to pin A6. A 4.7k resistor (R28) is connected between D+ and GND.
- SHIELD:** Connected to pin S1.
- GND:** Connected to pins A1 and B8.

FT232RL Module (U11) Details:

- Power:** VCC (pin 15) is connected to +3.3V. RESET (pin 14) is connected to +3.3V. GND (pins 6, 16) is connected to GND.
- UART:** TXD (pin 20) is connected to USART1_RX. RXD (pin 19) is connected to USART1_TX.
- Other Pins:** RYS (pin 4), CTS (pin 9), DTR (pin 1), DSR (pin 8), DCD (pin 5), RT (pin 3), CBUS0 (pin 18), CBUS14 (pin 17), CBUS2 (pin 10), and CBUS3 (pin 19) are connected to GND.
- LEDs:** D8 LED is connected to pin 18 through a 470 ohm resistor (R14). D9 LED is connected to pin 17 through a 470 ohm resistor (R29).

The diagram illustrates the connection of three components: STLink, Encoder, and J14. Each component is connected to a +3.3V power source and a GND (ground) connection. The STLink component has two pins labeled SYS_SWCLK and SYS_SWDIO. The Encoder component has two pins labeled TIM1_CH1 and TIM1_CH2. The J14 component has two pins labeled TIM3_CH1 and TIM3_CH2. The connections are labeled with pin numbers and component names.

The diagram illustrates three separate LED circuits. Each circuit consists of an LED connected in series with a resistor (R8 for Red, R9 for Green, R10 for Blue) to a GND pin. The LEDs are labeled D5 (Red), D6 (Green), and D7 (Blue). The resistors are labeled R8 (1k), R9 (1k), and R10 (1k). The LEDs are connected to the positive supply rail, and the resistors are connected to the microcontroller pins.